ARAB YOUTH SPACE HACKATHON 2026 · CHALLENGE 813 · FINAL

Blue Carbon Guardian

Tide-aware satellite monitoring for Gulf mangroves — from alert to evidence.

Team Blue Athar · الأثر الأزرق

Theme: Ecosystem Health, Biodiversity & Blue Carbon

91 stands
Abu Dhabi + Tarut Bay

5 confirmed
real conversions, 0.3–0.5 m

73 % vs 13 %
partial loss caught vs chance

≈1
false alarm / stand-year



Nahla Nabil · Bahrain

THE TEAM

One engineer, every role — delivered in the open



Nahla Nabil

Team lead · AI engineer · IEEE researcher · Bahrain

EDUCATION

BSc Artificial Intelligence · Arab Open University, Bahrain · final semester · First-Class Honours standing

RELEVANT EXPERTISE

- 2 IEEE papers: federated learning for industrial IoT (ICCA 2025); random-forest audit of GCC sustainability indicators (ICAISET 2026)
- AI & LLM product engineering (Doo Bahrain; FlyRank backend AI)
- Machine learning, computer vision (deepfake detection), Python geospatial stack

COMMUNITY

NASA Space Apps Challenge local lead (Manama): organising and mentoring space-data teams

ROLES & RESPONSIBILITIES IN THIS PROJECT

Project lead & product owner
problem framing, user needs, scope, every claim on these slides

Earth-observation data engineer
Sentinel-2 series (778 dates), EnMAP reader, WorldCover stands, sub-metre checks

Modelling & statistics
tide-aware model, stand + cell alert rule, calibration against chance

Validation & scientific integrity
real events, simulations, unseen stands, a withdrawn claim

Product & delivery
dashboard, website, site reports, alert API concept, notebook

Business & impact
customers, pricing hypotheses, unit economics, carbon with ranges

The gap we will close in incubation
a mangrove ecologist as mentor · a restoration or agency partner for site records · a GIS analyst for scale-out

THE PROBLEM

Restoration is scaling faster than monitoring

100 M

mangroves pledged by the UAE by 2030

COP26, 2021 · Abu Dhabi Mangrove Initiative, 2022

100 M+

mangroves pledged by Saudi Arabia by 2030

Saudi Green Initiative

Feb 2026

first GCC mangrove monitoring guide

EAD + IUCN Mangrove Specialist Group

TODAY

Periodic field surveys and one-off studies; a 2025 regional review calls Gulf monitoring **“uneven”**. No standing alert for each stand, no independent check of carbon claims.

THIS IS HAPPENING NOW · STAND 5, ABU DHABI



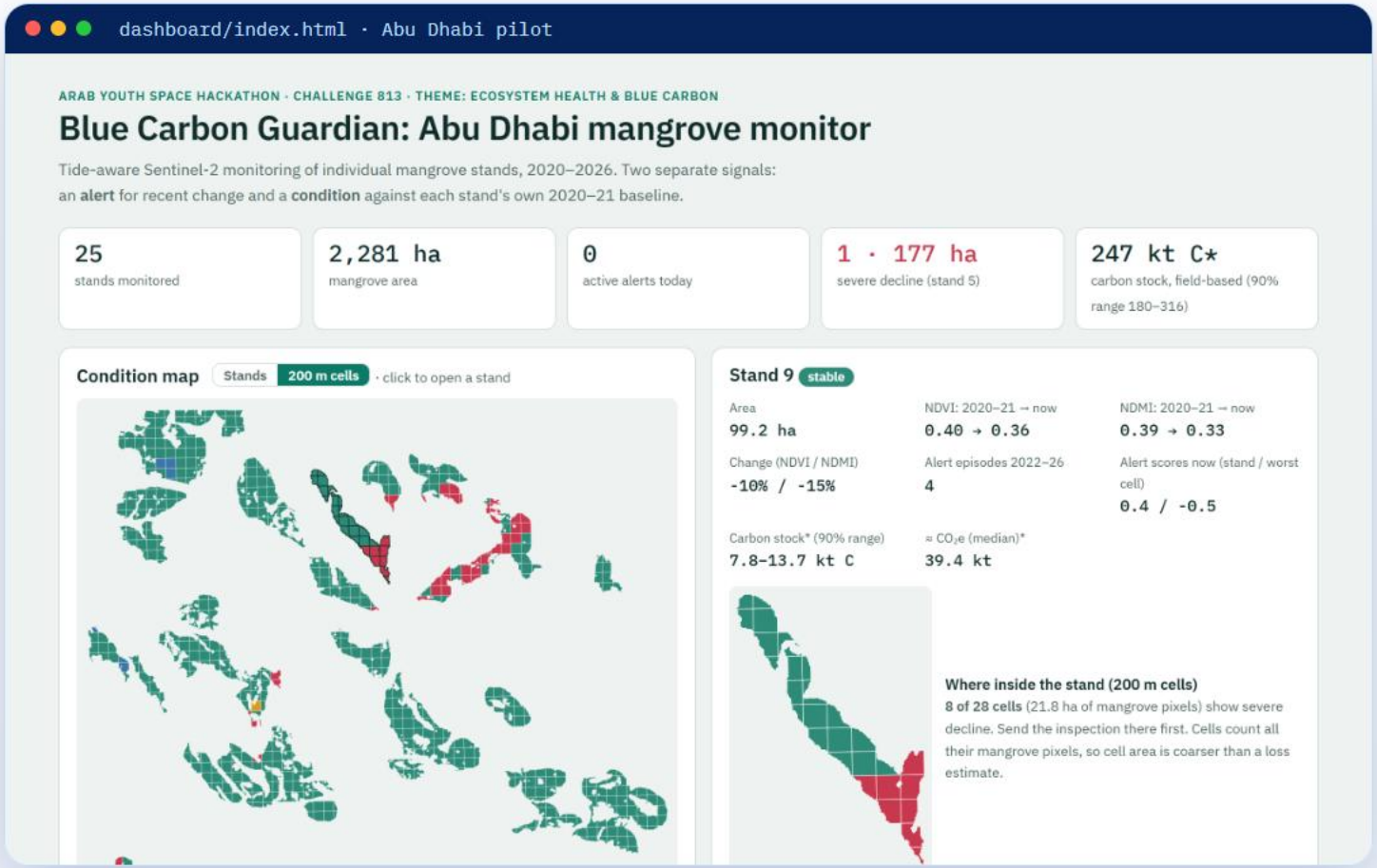
Sentinel-2 true colour, contains modified Copernicus Sentinel data. Our alert began on 25 Oct 2022, during the works.

SDG 13 SDG 14 SDG 15

Sources: ead.gov.ae · admangroves.ae · spa.gov.sa · Frontiers in Marine Science 2025 (doi:10.3389/fmars.2025.1695426)

EXAMPLE OUTPUT · THE PRODUCT

From signal to decision, in one screen



One alert
stand average OR any 200 m cell, ~1 false alarm per stand-year

Where to look
a 200 m cell map for the inspector

Condition
last 12 months vs the stand's own 2020–21 baseline

Carbon at stake
stock with a range, from field data

Site report
one printable page per stand; every sentence traces back to a number

dashboard · site reports · JSON alert API (concept)

live: nahla-nabil.github.io/blue-carbon-guardian

BUSINESS USE CASE & END USERS

Who uses it, and for which decision

Environment agencies

EAD, MOCCAE, Saudi NCVC

THEIR DECISION

Which stands do we inspect this month? Is a protected stand intact?

Today: Periodic field surveys, one-off studies

WITH BLUE CARBON GUARDIAN

Monthly alert + 200 m cell map: inspect only the flagged cells

Planting & carbon-project developers

ADNOC, Red Sea Global, carbon developers

THEIR DECISION

Is the carbon stock we report still there?

Today: Field plots and drone surveys, campaign by campaign

WITH BLUE CARBON GUARDIAN

Condition + carbon stock with a range per stand, a site report for verification

Coastal developers & regulators

EIA consultants, permit offices

THEIR DECISION

Did construction stay outside the mangrove line?

Today: Comparing two dated images by eye

WITH BLUE CARBON GUARDIAN

An alert when works enter mangrove cells, with dated evidence

New scene → every ~5 days

Alert → ~1 false alarm / stand-yr

Cell map → where to look

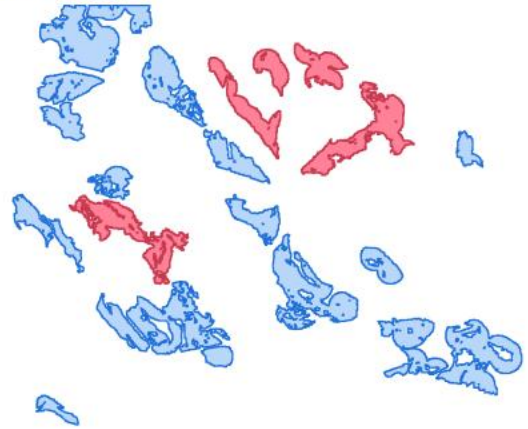
Inspection → only flagged cells

Site report → carbon at stake

DATA & STUDY AREA, INCL. HYPERSPECTRAL

91 mangrove stands, six years of free imagery

PILOT: 25 STANDS · 2,281 HA · ABU DHABI



Outlines: WorldCover 2021 patches ≥ 3 ha. Red: the five confirmed conversions. **+66 unseen stands** (Abu Dhabi N/W, Saudi Tarut Bay) test transfer.

Sentinel-2 L2A
ESA / Copernicus · time series, 10–20 m
free · 778 dates

EnMAP
DLR · 224 bands · independent hyperspectral check
free · 2 scenes

ESA WorldCover 2021
10 m land cover · stand outlines
CC BY 4.0

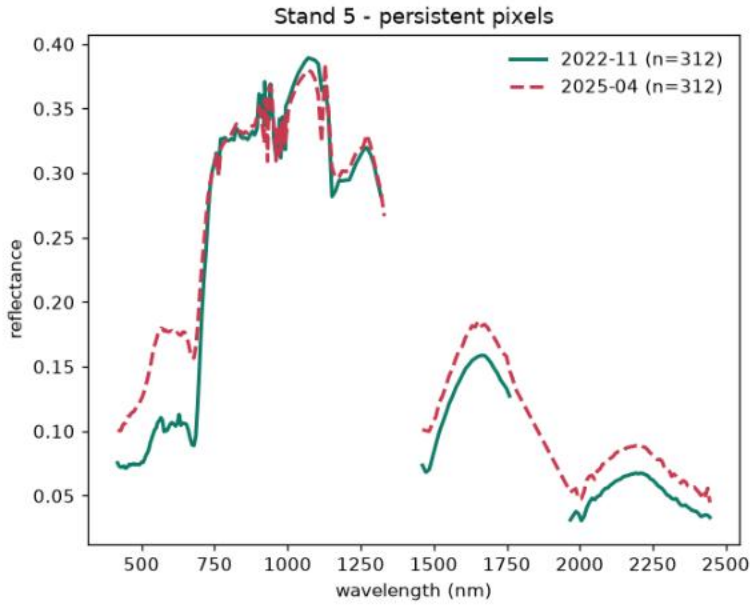
Field carbon plots
Schile et al. 2016 · carbon density
CC0 · 24 plots

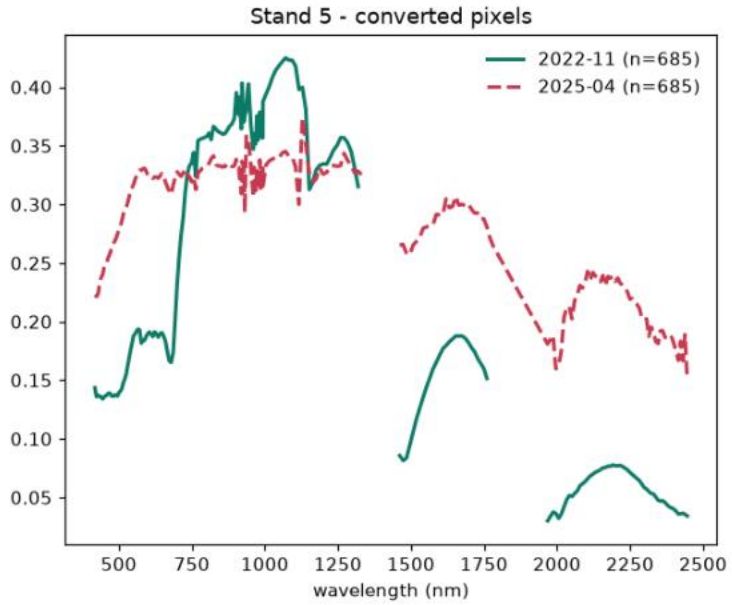
Sub-metre captures
WorldView-2/3, Legion-1 · confirming conversions
viewed only (Esri Wayback)

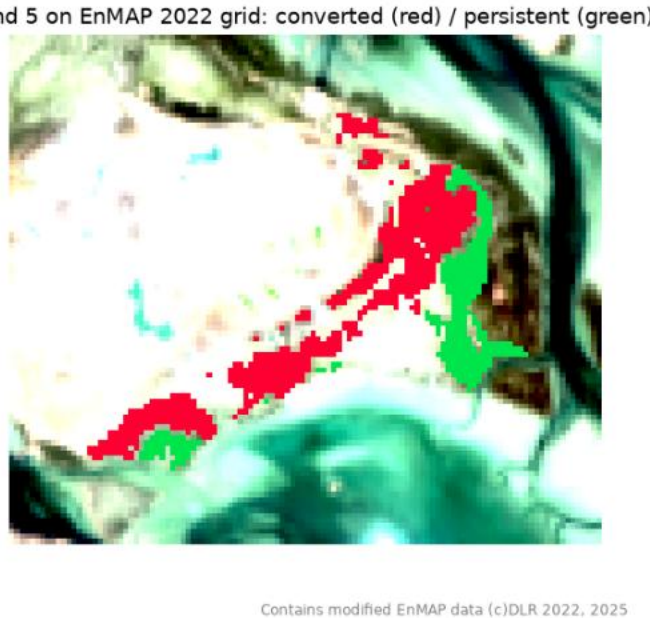


VALIDATION 2 · HYPERSPECTRAL (ENMAP)

An independent check, and a material fingerprint







Contains modified EnMAP data (c)DLR 2022, 2025

r = 0.84–0.99
EnMAP vs our Sentinel-2 indices, same dates, 25 and 14 stands

62 vs 26 ha
stand 5 converted: EnMAP pixels vs our conservative Sentinel-2 flag

F1 0.74 vs 0.72
full spectrum vs Sentinel-2-like bands as a classifier: no real gain, and we say so

Converted pixels change from a vegetation spectrum to a bright mineral-like one (sand and fill): reclamation, not dieback. Contains modified EnMAP data © DLR 2022, 2025.

SPACE DATA · READY FOR SATELLITE 813

Sentinel-2 says when. Satellite 813 says what.

Sensor	Bands	Range	Pixel	Revisit	Role in the product
Sentinel-2	13	443–2190 nm	10–20 m	~5 days	**when**: the alert, every pass
EnMAP (used)	224	420–2450 nm	30 m	~27 days	**what**: tested on stand 5
Satellite 813 (next)	~205	~400–1700 nm	~20 m	on demand	**what, sharper**: tasked on the flagged cells

1 · Sentinel-2 alert
stand or 200 m cell below its threshold

→

2 · Task Satellite 813
only over the flagged cells, on demand

→

3 · Spectral diagnosis
vegetation or sand and fill: evidence pack

Every index the monitor uses (NDVI, red edge, SWIR moisture at 1610 nm, wetness) sits inside 813's ~400–1700 nm range, at 20 m instead of EnMAP's 30 m. **Planned, not yet tested:** 813 scenes were not available in the proof of concept.

APPROACH · WORKFLOW

From satellite pixels to an inspection decision

1 · Stands & cells

WorldCover 2021 patches ≥ 3 ha, cut into 200 m cells; interior pixels only

2 · Sentinel-2 series

778 dates 2020–26; cloud mask; NDVI, red edge, SWIR moisture, wetness

3 · Tide-aware model

robust trend + season + tide terms, refitted every month

4 · Alert & condition

stand ≤ −1.5σ OR any cell ≤ −3.5σ; last 12 months vs 2020–21

5 · Decision layer

cell map, carbon at stake, dashboard, site reports

CHECKS BUILT IN, INDEPENDENT OF THE MODEL

EnMAP hyperspectral
same dates, r = 0.84–0.99

Sub-metre captures
5 conversions confirmed

Simulated losses
every rate vs its chance rate

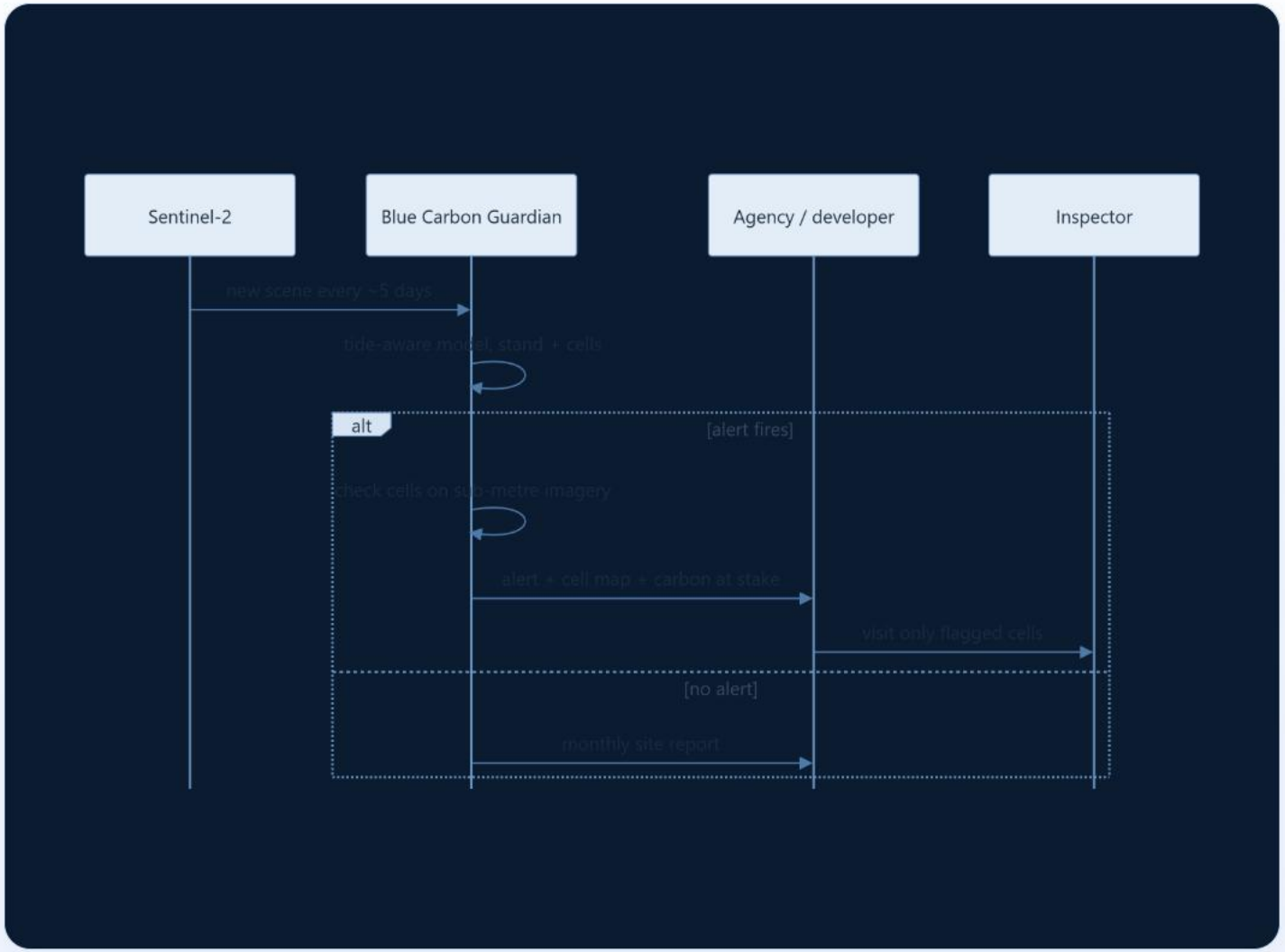
65 unseen stands
0.79 false alarms / stand-yr

PROBLEM STATEMENT

"We monitor mangrove condition and carbon stock in Abu Dhabi, 2020–2026, so that authorities and developers can decide where to inspect and verify carbon claims."

DELIVERY · HOW A USER RECEIVES IT

One alert, one message, one targeted visit



Every ~5 days

a new Sentinel-2 scene runs through the tide-aware model, per stand and per 200 m cell

If an alert fires

sub-metre check of the flagged cells, then an email or JSON alert with the cell map and the carbon at stake

If nothing changes

a monthly site report: condition, trend and carbon stock

TECHNICAL INNOVATION 1

The tide looks like tree loss — until you correct for it

WHY NAIVE MONITORS FAIL ON GULF TIDAL FLATS

same trees, same day of the year

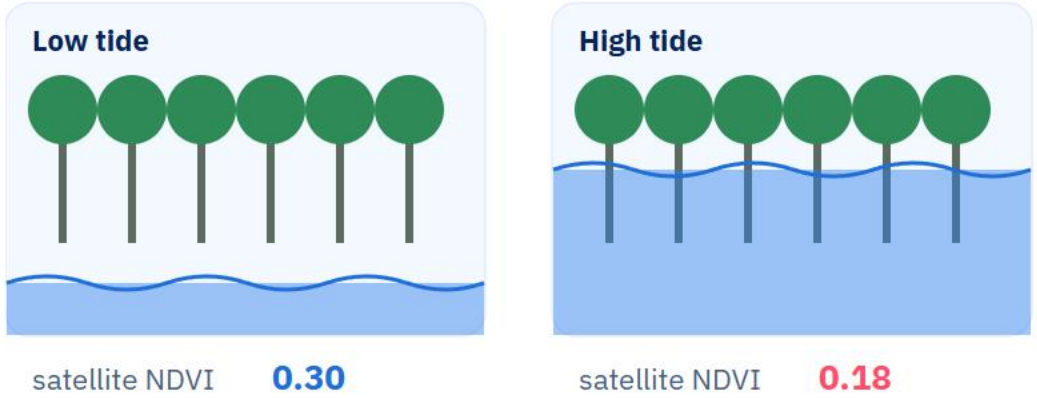
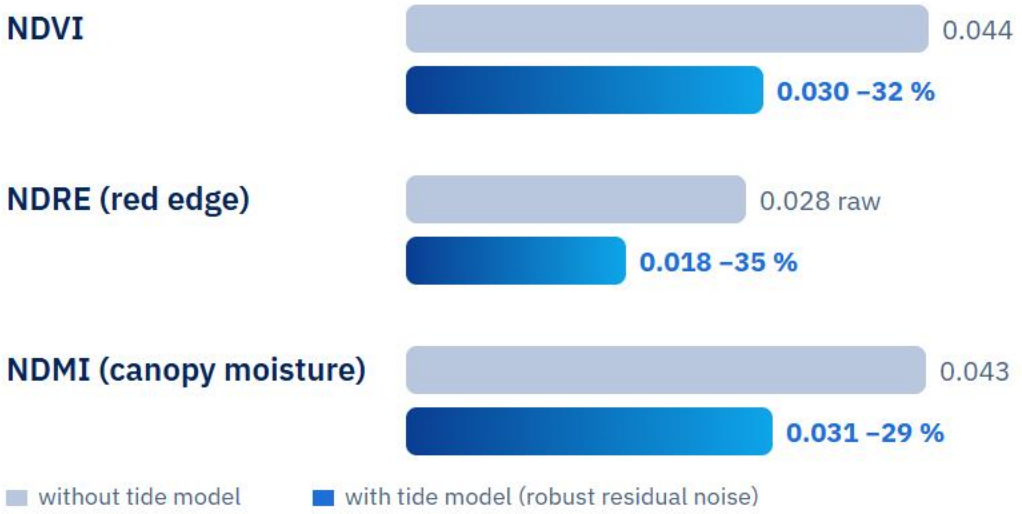


Illustration. Measured on our stands: index noise vs wetness, **r ≈ −0.9**.

DATE-TO-DATE NOISE, 25 STANDS, 2020–2026



We regress each index on trend, season **and tidal state** before flagging anything: **29–35 % less noise** on every index. Same alert rule, same stands: textbook NDVI-only **1.61** false alarms per stand-year, ours **0.96** (40 % fewer).

TECHNICAL INNOVATION 2

A stand average hides partial loss. 200 m cells do not.



Stand 5, winter 2022–23 (real data): 200 m cells turn red while the stand average barely moves. Contains modified Copernicus Sentinel data.

8 of 28

cells of stand 9 in severe decline while its average reads **“stable”**

3 of 5

real conversions found **only** by the cells (stands 12, 18, 6)

ONE ALERT RULE

stand ≤ −1.5σ **OR** any cell ≤ −3.5σ

calibrated together: 0.96 false-alarm episodes per stand-year

Partial loss caught within 60 days (simulated)

73 % stand + cells

25 % stand average alone

13 % by chance

INNOVATION & DIFFERENTIATION

How we differ from what exists today

	Field surveys	One-off satellite maps	Global Mangrove Watch alerts	Blue Carbon Guardian
How often	campaigns	yearly or less	monthly	**every ~5 days**
Corrects for the tide	n/a	not usually	not documented	**✓ tide model**
Says where inside a stand	✓	—	—	**✓ 200 m cells**
Published false-alarm rate	n/a	n/a	accuracy only	**✓ ≈ 1 per stand-year**
Carbon at stake per alert	—	—	—	**✓ field-based**
Gulf coverage	✓	✓	not confirmed	**✓ Abu Dhabi + Tarut Bay**

Complementary, not competing: Global Mangrove Watch flags loss worldwide; we add the tide model, the 200 m cells, a measured false-alarm rate and the carbon at stake — built and tested for the Gulf.

VALIDATION 1 · REAL EVENTS

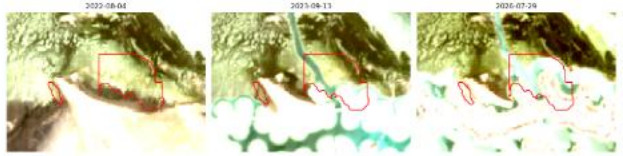
Five real conversions, confirmed on sub-metre imagery



Stand 5 · reclamation with finger canals



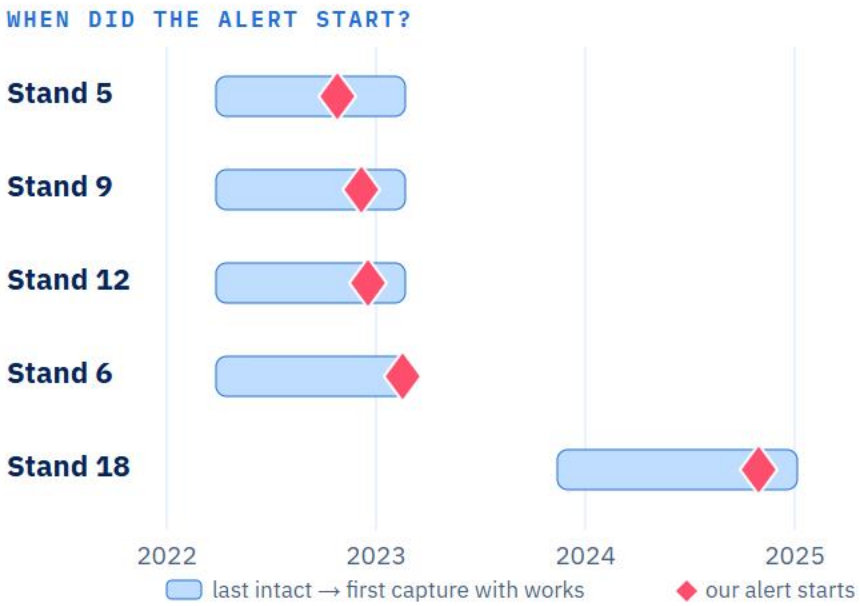
Stand 9 · canal development takes the south-east



Stand 12 · found only by the cells



Stand 18 · found only by the cells, 2025



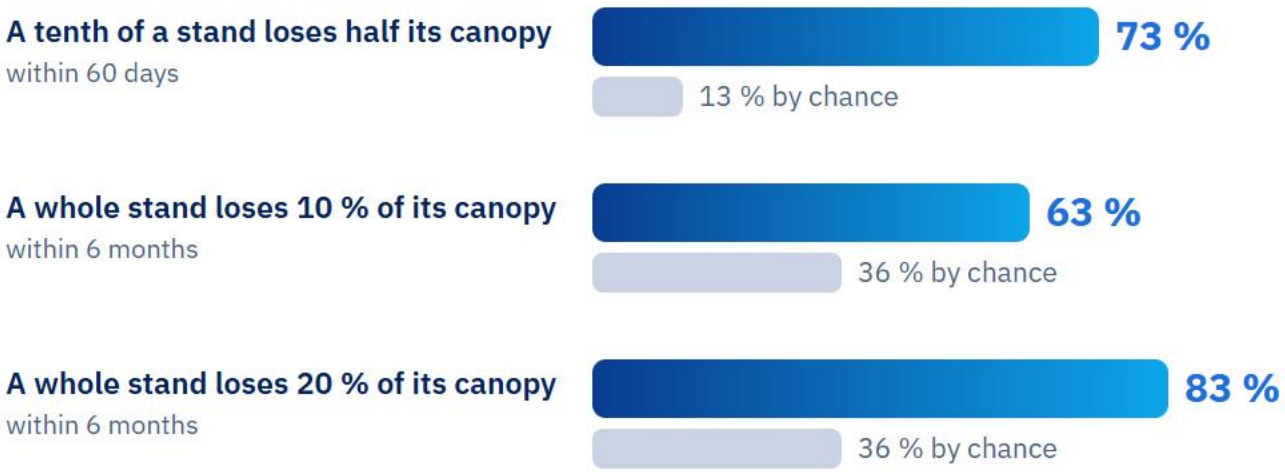
Dated 0.3–0.5 m captures (WorldView-2/3, Legion-1) confirm all five, including stand 6 where a lagoon replaced part of the stand. At every site the alert began **between the last intact and the first disturbed capture**: detection of works as they happen, not advance warning.

VALIDATION 2 · AGAINST CHANCE, OUT OF SAMPLE, SECOND SENSOR

Every rate next to its chance rate

ARTIFICIAL LOSS INJECTED INTO REAL STANDS WITH NO KNOWN CHANGE

Counts only a **new** alert after the loss starts; the same test with no loss gives the chance rate. ~160–240 trials per row.



FALSE ALARMS · CALIBRATED, 25 STANDS

0.96

episodes per stand-year

UNSEEN · 65 STANDS, THRESHOLDS FIXED

0.79

per stand-year, 305 stand-years, incl. Saudi Tarut Bay

ENMAP HYPERSPECTRAL · SAME DATES

r = 0.84–0.99

vs our Sentinel-2 indices; stand-5 loss 62 ha vs our conservative 26 ha

⚠ **What we do not claim:** advance warning, field verification, species discrimination, or that hyperspectral classifies better (F1 0.74 vs 0.72). We withdrew our own first test (58 % / 90 %) when we found it had no chance baseline.

IMPACT & BUSINESS VIABILITY

Carbon at stake, and who pays to protect it

108 t C/ha
field density near the stands (90 % CI 79–138), 82 % of it in the soil

247 kt C
pilot stock, 2,281 ha (range 180–316) ≈ 907 kt CO_2e

2.8–6.7 kt C
stock on the 26–62 ha lost at stand 5: what one alert puts on the table

SERVICE TIERS (PRICES ARE PILOT HYPOTHESES)

Monitor
$12k
per site / yr
alert, cell map, condition, carbon, monthly reports

Evidence
$25k
per site / yr
+ sub-metre check of every alert, audit pack

Development watch
$8k
per project / yr
for coastal developers and EIA consultants

Cost to serve ≈ **$3.3k** / site / yr: compute < $5 (measured), sub-metre checks on alerts $500–750, analyst $2k (assumption).

Customers: agencies (EAD, MOCCAE, NCVC) · planting programmes (ADNOC 10 M, Red Sea Global 50 M) · carbon developers · EIA consultants

COMPETITION

Global Mangrove Watch sends free monthly loss alerts (Gulf coverage not confirmed). We are complementary: tide-aware, calibrated, stand + cell, carbon and reports.

THE HONEST MARKET

Gulf + Red Sea site subscriptions ≈ $0.2–0.4 M / yr. The real prize: a **national MRV contract** or a monitoring layer on **gIQ**.

SDG 13 · SDG 14 · SDG 15

BONUS CRITERIA · BEYOND THE PROOF OF CONCEPT

Reproducible today, built to scale tomorrow

BONUS · DOCUMENTATION, EVIDENCE & REPRODUCIBILITY

Public repository, MIT licence
code, notebook, figures, data provenance and licences

Re-runs end to end in ~24 s
notebook with outputs committed; pinned requirements; example input + output

Clean-machine test passed
fresh environment, same numbers, 0 errors

Every number traceable
each figure on these slides comes from a script and a written summary

Live website + site reports
anyone can open the monitor and check a stand

`github.com/Nahla-Nabil/blue-carbon-guardian`

BONUS · SCALABILITY BEYOND THE POC

Proven now

91 stands, 2 countries, 2 Sentinel-2 tiles · 65 unseen stands: 0.79 false alarms / stand-yr · compute < $5 per site per year · backfill: 22–73 min per block

Incubation

Satellite 813 in the spectral step · deployment on gIQ · a pilot customer + site records · more EnMAP dates per season

Scale

Gulf ≈ 165 km² + Red Sea ≈ 200 km² of mangrove · sub-metre data for small stands (Bahrain) · national MRV · platform licensing

Mangrove areas: Frontiers in Marine Science 2025 (doi:10.3389/fmars.2025.1695426). Compute measured on our runs; prices and analyst time are assumptions.

TEAM BLUE ATHAR · الأثر الأزرق

Every lost mangrove leaves a trace. We find it from space.

Tide-aware
29–35 % less noise

5 of 5
real conversions alerted during the works, 3 only by cells

0.79
false alarms / stand-yr on unseen stands

Open
public code, re-runs in ~24 s

Our ask: incubation to run a pilot with a Gulf agency or planting programme, mentor access on hyperspectral reading of mangroves, and Satellite 813 scenes when available.

nahla-nabil.github.io/blue-carbon-guardian · github.com/Nahla-Nabil/blue-carbon-guardian · Thank you



Thank you

شكراً للإمارات

To the **United Arab Emirates** — for leading the region in protecting its mangroves, and for opening space to Arab youth.

UAE Space Agency · National Space Academy · Space42 · the mentors and the judges

From Bahrain, with gratitude — Nahla Nabil · Team Blue Athar

BACKUP · FOR QUESTIONS

The textbook approach against ours, same alert rule

Configuration	Indices	Tide model	False alarms per stand-year	Real conversions caught
A · textbook	NDVI	no	**1.61**	5 of 5
B	NDVI	yes	**1.04**	5 of 5
C	NDVI + red edge + SWIR moisture	no	**1.36**	5 of 5
D · ours	NDVI + red edge + SWIR moisture	yes	**0.96**	5 of 5

40 % fewer false alarms

20 stands with no known change, 93.9 stand-years, 2022–2026. About 16 fewer unnecessary inspections a year for 25 stands.

The tide model is most of the gain

NDVI alone 1.61 → with tide terms 1.04; the extra indices bring it to 0.96.

Honest limit

Thresholds calibrated for our monitor; all four catch the five conversions. The difference is the false-alarm cost, not detection.

BACKUP · FOR QUESTIONS

Where it breaks — and what we do not claim

Detection, not early warning

Alerts fire as works happen. We never claim to see stress before it is visible.

10 m pixels

Patches under ~0.6 ha and young plantings are missed, Bahrain's small stands included.

Imagery, not site visits

Conversions confirmed on dated sub-metre captures, not yet on permits or field records.

Simulated losses

Canopy-cover loss, not leaf stress; every rate next to its chance rate.

Hyperspectral

EnMAP confirms our indices but was not a better classifier (F1 0.74 vs 0.72).

Carbon

Area × field density from four sites (2013–14): a stock with a range, never a sequestration rate.

Rule we kept: no detection rate without its chance rate; no claim of advance warning, field verification or species discrimination.